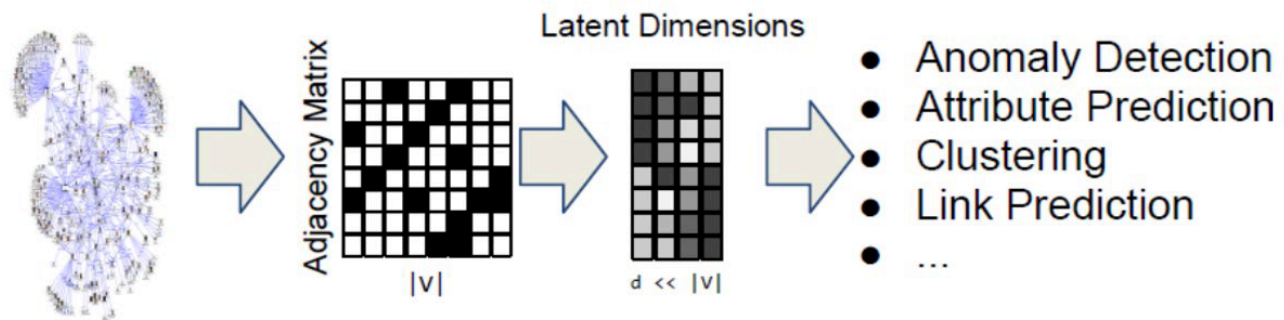


L14: Applications of Node Embeddings



Images from Jure Leskovec, "Representation Learning on Networks" tutorial, WWW 2018

How can we use node embeddings to answer important questions about the original network?

If you are familiar with machine learning, you know that some classical ML tasks are to perform classification, clustering, regression, anomaly detection, feature learning, etc. All of these tasks have their counterparts in network analysis.

For instance, ML classifiers (based on algorithms such as neural nets, decision trees, SVMs, etc) can be directly applicable to **classify the nodes of a network in different types** (e.g., male versus female, humans versus bots, political affiliation). To perform this task we need a training dataset that consists of pairs $(\mathbf{z}_v, c_v)$, where $\mathbf{z}_v$ is the embedding vector of node v and c_v is the class (or type) of that node, at least for some nodes of the graph.

Another ML application on networks is **link prediction**. As we have discussed before, network data are often noisy and we may have both false positives (spurious edges) and false negatives (missing edges). If we have computed node embeddings based on a (potentially noisy) estimate of the adjacency matrix, we can then train a binary classifier to predict whether any two given nodes are connected or not. Again, we need a training dataset that consists of pairs of nodes for which we are confident that they are either connected -- or not connected. That classifier can then be applied on other pairs of nodes for which we do not have high-quality data.

A third application is **community detection**. Even though we have already discussed several algorithms to solve that problem, the use of node embeddings allows us to apply any *clustering algorithm* on those vectors. If we use an embedding scheme that considers the network distance between any two nodes, each resulting cluster will consist of a group of nodes that are within a short network distance from each other. The ML literature includes 100s of clustering algorithms, and so using this approach for community detection provides us with a very large "toolbox".

A fourth application is the **visualization of networks**, relying on the dimensionality reduction provided by node embeddings. Suppose for instance that the node embeddings are two-dimensional. Then, we can represent the nodes of the network in the Euclidean plane, which is much easier for humans to visualize and understand. Nodes with shorter network distance will have more similar embeddings.

There are many other applications of node embeddings in the analysis of networks -- and the list keeps growing. As we will see in the next page, however, the use of shallow embeddings has some fundamental limitations. For this reason, we will discuss next the use of deep neural networks to represent graphs.